Food and Agriculture Organization of the United Nations

Discover

English

Home > News and Events > News > Detail

# KOFAP - Korea FAO Sustainable and Innovative Fisheries and Aquaculture Programme

## Boosting biosecurity in Peru’s shrimp farming industry for sustainable livelihoods

**03/06/2025**

The farming of whiteleg shrimp (*Penaeus vannamei*) is an important activity in Peru’s economy, with total production volume of 43 000 tonnes in 2023, according to FAO.

The shrimp farming industry is especially vibrant in the Tumbes region near the border with Ecuador. Here, a total of 34 shrimp farms - 28 semi-intensive and six intensive - account for approximately 85 percent of the country's production. Tumbes also has six processing plants, some artisanal and some industrial.

The sector provides significant employment opportunities, particularly for women, who are actively involved in shrimp processing. In 2023, the shrimp value chain in the Tumbes area provided 1600 full-time jobs.

Also known as Pacific white shrimp or king prawn, this nutritious crustacean can be cultivated in both marine and brackish water environments and is a key national export product in both frozen and processed form. In 2023, the shrimp industry generated export revenues valued at approximately USD 263 million, according to the Ministry of Production (PRODUCE).

## A challenge to be overcome

Like many economically important aquatic animals, farmed shrimp can be vulnerable to contagious diseases that spread quickly and cause extensive mortality, leading to economic losses and affecting livelihoods all along the value chain.

These include the lethal White Spot Syndrome Virus (WSSV) and bacterial infections such as necrotizing hepatopancreatitis (NHP) and acute hepatopancreatic necrosis disease (AHPND), all of which have been detected and reported by Peru's National Fisheries and Aquaculture Health Authority (SANIPES).

Recognizing these challenges, SANIPES partnered with the Food and Agriculture Organization of the United Nations (FAO) to build the capacities of farmers, local authorities, technicians, and laboratory personnel through practical training workshops on the Progressive Management Pathway for Aquaculture Biosecurity (PMP/AB).

Participants gained hands-on skills in early identification of shrimp diseases by observing animal behaviour, correct tissue sampling, and performing simple yet effective diagnostic tests in the field.

"I had an outbreak of NHP on my farm, and it affected a lot of my stock," says Shirley Terrones, who is in charge of quality control on a 430-hectare shrimp farm with 81 ponds in Tumbes.

"I attended the workshop because I wanted to know how to prevent this from happening again. It was quite useful because I learned to spot the presence of the disease quickly and apply control measures before it spreads. This way I can avoid economic losses in future,” explains Ms Terrones.

## What is the PMP/AB and why is it important?

The PMP/AB is a framework for farmed aquatic animal and plant health management and disease control.

Expressly included in the FAO Guidelines for Sustainable Aquaculture, its aim is to improve biosecurity practices to support food security and nutrition, rural livelihoods and environmental sustainability.

It includes guidance on developing regional and national aquatic organism health strategies, including public-private partnerships to establish disease control systems and the co-management approach to enhance community-led biosecurity and environmental monitoring.

“By adopting the PMP/AB, shrimp farmers and local authorities can systematically identify biosecurity gaps and implement practical solutions tailored to their local context, ultimately reducing disease outbreaks and safeguarding production,” says Muriel Gomez-Sanchez, Director of the Department of Health and Safety at SANIPES.

## About the project

The Smart and Sustainable Aquaculture through Effective Biosecurity and Digital Technology (SAB) project is funded by the Republic of Korea and implemented in Peru and Sri Lanka.

SAB works to integrate digital technologies such as mobile applications, real-time monitoring sensors and data-driven early warning systems into aquaculture biosecurity management, equipping farmers and producers to swiftly detect and respond to disease threats.

Through SAB, FAO and SANIPES are working together to maintain biosecurity, ensure consumer safety, protect Peru's access to international markets, and support authorities to develop a national aquatic organism health strategy.

By empowering local communities and producers with technical skills and modern digital solutions, Peru will be better positioned to expand its shrimp industry and secure sustainable livelihoods for its people.


**PHOTO GALLERY**

Course "Surveillance and biosecurity of shrimp diseases"